

Revision · Surface area

Cambridge insert: unfold the solid into a net, and the surface area is just an area sum.

LESSON 66

1 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 14.2

LESSON CLOCK

5'

14'

12'

5'

4'

Unfold a cardboard box	5 min
Nets and cuboids	14 min
Cylinders	12 min
Practical problems	5 min
Homework	4 min

LEARNING OBJECTIVES

- Draw the net of a prism or a cylinder.
- Find the surface area of a cuboid, a prism and a cylinder.
- Distinguish curved surface area from total surface area.
- Solve practical problems of painting, wrapping and covering.

TEXTBOOK

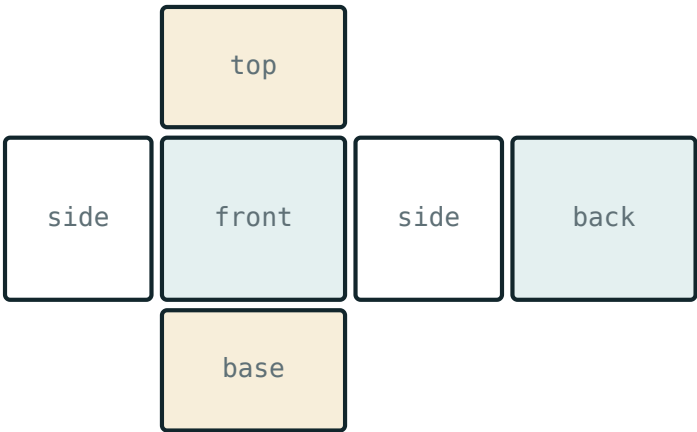
National	Annual revision
Cambridge	Learner's Book pp. 327–334
Workbook	Workbook 14.2

01

Explanation

Unfold it

Cut a solid open along its edges and lay it flat. The result is its **net**, and the surface area is simply the total area of the net — no new formula, just the areas you already know.



surface area = the area of the whole net

The net of a cylinder: two circles and one rectangle whose width is the circumference.

THE STANDARD RESULTS

cuboid: $S = 2(ab + bc + ac)$ | cube: $S = 6a^2$

cylinder: $S = 2\pi r^2 + 2\pi rh$ (two circles plus the curved surface)

The curved surface of a cylinder unrolls into a **rectangle**: its height is h , and its width is the circumference $2\pi r$. That is where $2\pi rh$ comes from — worth deriving once rather than memorising.

For any prism: $S = 2 \cdot S_{\text{base}} + P_{\text{base}} \cdot h$ — two ends plus the band round the side.

Reading the question

WHICH FACES ARE ACTUALLY THERE?

An **open-topped** box has five faces, not six. A pipe has no ends at all — only the curved surface. A tin needs both ends; a label needs only the curved part. Count the faces the question really describes before writing any formula.

Surface area is measured in square units even though the object is three-dimensional — it is, after all, the area of a flat net. Volume takes cubic units. Mixing the two is the commonest error in this topic.

02 Worked examples

EXAMPLE 1 Find the surface area of a cuboid 4 cm by 5 cm by 6 cm.

① $4 \cdot 5 = 20$, twice
The top and bottom.

② $5 \cdot 6 = 30$, twice
Two sides.

③ $4 \cdot 6 = 24$, twice
The other two.

④ $2(20 + 30 + 24) = 148 \text{ cm}^2$

ANSWER

148 cm^2

EXAMPLE 2 A cylinder has radius 5 cm and height 12 cm. Find its curved surface area and its total surface area.

① $\text{curved} = 2\pi \cdot 5 \cdot 12 = 120\pi$
The unrolled rectangle.

② $\text{two circles} = 2\pi \cdot 25 = 50\pi$

③ $\text{total} = 170\pi \approx 534 \text{ cm}^2$

ANSWER

Curved 120π , total $170\pi \approx 534 \text{ cm}^2$

EXAMPLE 3 An open-topped box is 30 cm by 20 cm by 15 cm high. Find the area of card needed.

- 1 $\text{base} = 30 \cdot 20 = 600$
One base only — no lid.
- 2 $\text{two sides} = 2 \cdot 30 \cdot 15 = 900$
- 3 $\text{two ends} = 2 \cdot 20 \cdot 15 = 600$
- 4 $600 + 900 + 600 = 2100\text{ cm}^2$

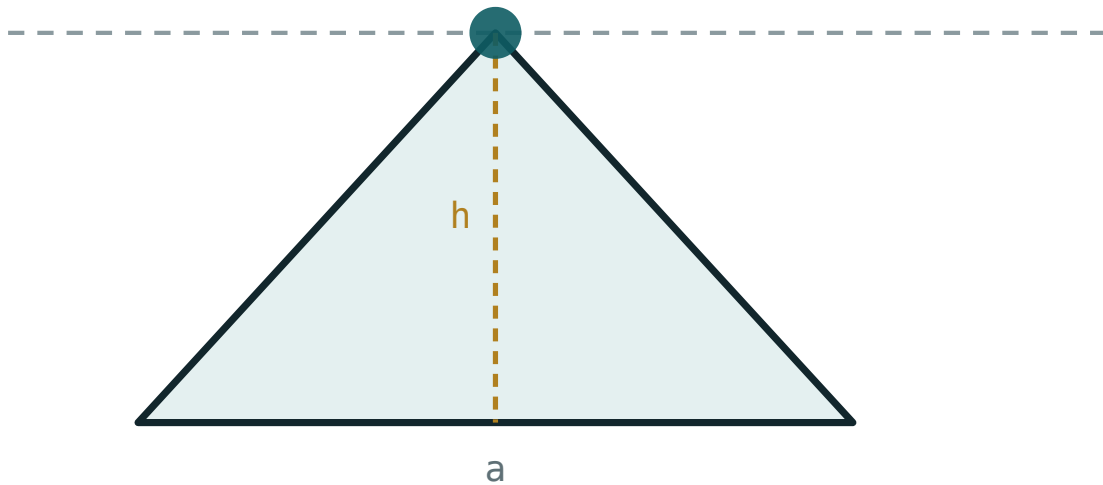
ANSWER

2100 cm^2

03 Interactive model

Change the dimensions and total the faces of the net as you go.

One face of the net

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Surface area	Sirt yuzasi	Площадь поверхности
Net	Yoyilma	Развёртка
Face	Yoq	Грань
Edge	Qirra	Ребро
Curved surface	Yon sirt	Боковая поверхность
Total surface area	To'liq sirt yuzasi	Полная поверхность
Base area	Asos yuzasi	Площадь основания
Open-topped	Ustki qopqoqsiz	Без крышки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cube of side 5 has surface area:

A cuboid $2 \times 3 \times 4$ has surface area:

The curved surface of a cylinder is:

An open-topped box has how many faces?

4

5

6

8

Surface area is measured in:

cubic units

square units

linear units

litres

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the surface area of a cube of side 3.

ANSWER 54

2. Find the surface area of a cuboid $2 \times 4 \times 5$.

ANSWER 76

3. Find the curved surface area of a cylinder with $r = 2$, $h = 6$ (in terms of π).

ANSWER 24π

4. Find the total surface area of that cylinder.

ANSWER 32π

5. A cube has surface area 96. Find its side.

ANSWER 4

6. How many faces has a triangular prism?

ANSWER 5

7. Find the surface area of a cuboid $1 \times 1 \times 10$.

ANSWER 42

Medium · the standard question

7 PROBLEMS

1. Find the surface area of a cuboid $6 \times 8 \times 10$.

ANSWER 376

2. Find the total surface area of a cylinder with $r = 4$, $h = 9$.

ANSWER $104\pi \approx 327$

3. An open-topped box is 40 by 25 by 20. Find the card needed.

ANSWER 3600 cm^2

4. A triangular prism has a 3–4–5 triangle cross-section and is 10 long. Find its surface area.

ANSWER $2 \cdot 6 + 12 \cdot 10 = 132$

5. A cylinder has curved surface area 60π and radius 5. Find its height.

ANSWER 6

6. A cube has volume 216. Find its surface area.

ANSWER 216

7. A label wraps the curved surface of a tin of radius 3.5 cm and height 10 cm. Find its area.

ANSWER $70\pi \approx 220 \text{ cm}^2$

Hard · stretch and reason

7 PROBLEMS

1. A cylinder has total surface area 100π and radius 5. Find its height.

ANSWER $50\pi + 10\pi h = 100\pi$, so $h = 5$

2. A room 5 m by 4 m by 3 m high is painted — walls and ceiling only. Find the area painted.

ANSWER $54 + 20 = 74 \text{ m}^2$

3. A cube is cut into eight identical smaller cubes. Compare the total surface area before and after.

ANSWER It doubles — each small cube has $\frac{1}{4}$ of the original face area but there are eight of them, so $8 \cdot 6 \cdot \frac{a^2}{4} = 12a^2$ against $6a^2$.

4. A pipe of radius 6 cm and length 2 m is open at both ends. Find its outer curved surface area.

ANSWER $2\pi \cdot 6 \cdot 200 = 2400\pi \approx 7540 \text{ cm}^2$

5. Two similar cylinders have radii 3 and 5. The smaller has surface area 90π . Find the larger.

ANSWER $k = \frac{5}{3}$, $k^2 = \frac{25}{9}$, so 250π

6. A cuboid has volume 120 cm^3 and a square base of side 4 cm. Find its surface area.

ANSWER $h = 7.5$, $S = 32 + 120 = 152 \text{ cm}^2$

7. Explain why doubling every length multiplies surface area by 4 but volume by 8.

ANSWER Surface area is a sum of products of two lengths, so it scales by k^2 ; volume is a product of three, so it scales by k^3 .

07 Homework

Homework — 5 problems

Cambridge Stage 9, Unit 14.2. Sketch the net before totalling.

1. Find the surface area of a cuboid $5 \times 7 \times 9$.

2. Find the total surface area of a cylinder with $r = 3$ and $h = 8$.
3. An open-topped box is 50 by 30 by 25. Find the card needed.
4. A cube has surface area 150 cm^2 . Find its side and its volume.
5. A label wraps a tin of radius 4 cm and height 12 cm. Find its area.